# William Miller

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## EDUCATION

**Tufts University** Medford, MA
*Bachelor of Science in Mechanical Engineering* *Expected: May 2028*
GPA: 3.98
**Relevant Courses:** Mechanics I & II, Thermal Fluid Systems I & II, Mechanical Design I & II, Electronics and Controls, Computer-Aided Product Design, Materials and Manufacturing, Numerical Methods

## ENGINEERING EXPERIENCE

**Tufts Solar Vehicle Project** Medford, MA
*Dynamics Lead* *Apr 2026 – Present*

- Lead a team of ~10 members on the vehicle's dynamics systems; reviewed past designs to identify what still needs to be designed and built
- Directing design of the brake system and pedal box, including braking calculations and a parking brake, and planning suspension and steering integration once the chassis is complete
- Mentor new members in SolidWorks CAD and FEA

*Team Member* *Sept 2025 – Apr 2026*

- Designed a roll cage in SolidWorks and developed a PVC prototype with 3D-printed elbow joints to validate geometry, then redesigned for manufacturability after identifying bending limitations with available tooling
- Running FEA in SolidWorks Simulation on the front hoop across 5 required load cases (combined 5g/4g/1.5g loading, 30° and 60° angled loading, lateral and rearward horizontal loading) to verify stresses remain within yield strength per competition safety regulations

**Chiesa Research Group (Superconductivity Lab)** Medford, MA
*Undergraduate Research Assistant* *June 2026 – Oct 2026*

- Co-designed and machined custom Instron grips (CAD, sizing for strength and fit, lathe work) to enable round-sample mechanical testing for an external industry client; grips sustained loads up to 40 kN with no sample slip
- Characterized and corrected measurement error in a video extensometer caused by cylindrical sample geometry and developed a modified camera-calibration procedure that reduced a ~5 mm gauge-length error to within 1 mm

## WORK EXPERIENCE

**Tufts Department of Physics** Medford, MA
*Learning Assistant* *Sept 2025 – Dec 2025*

- Selected by course instructor to guide students through complex physics problems during lectures and office hours, fostering discussion through targeted questioning and encouraging multiple solution approaches
- Provided weekly feedback to the instructor on student progress and common misconceptions

## PROJECTS

**Motor-Controlled Clock** In Progress

- Designing a 4-digit clock with an analog (mechanical) 7-segment display, where each digit is driven by 7 inverse cam pairs with a custom cam profile for each segment
- Programming stepper motor control in C++ on an Arduino to drive the mechanism
- Modeled the full mechanism in SolidWorks, including a gear train sized for stepper motor actuation, and 3D printed the majority of components

**Assistive Paint Dispenser** Apr 2026 – May 2026

- Designed an assistive paint dispenser in SolidWorks for a client with cerebral palsy, accommodating two paint tube sizes with an ergonomic 3D-printed handle and die and an aluminum frame manufactured via waterjet cutting
- Prototyped iteratively using laser-cut acrylic to validate geometry before final fabrication, and produced custom paint trays via vacuum forming to complete the accessible workspace solution
- Delivered the finished device to the client

## SKILLS

**CAD & Simulation:** SolidWorks (CSWA certification), SolidWorks Simulation (FEA), Onshape
**Programming & Analysis:** MATLAB, C++, Circuit Python (KB2040), Excel
**Manufacturing & Prototyping:** 3D printing, laser cutter, waterjet cutter, vacuum forming, manual mill, manual lathe (including threading)